Output-Trace Identifiability Gaps in LLM Agent Audit: Behavioral Equivalence Is Not Audit Equivalence

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Abstract

Two agent realizations can produce identical output-trace distributions yet differ maximally in residual decision relevance: one realization can have $\delta_{\text{act}} = 0$ (hidden state is dormant) while another has $\delta_{\text{act}} = H(A_t \mid \tilde{T}_t)$ (hidden state drives every action), with no output-only observable distinguishing them. We prove this Output-Trace Identifiability Gap as Theorem 1 and then close it with a dual-certificate protocol: a structural-access closer $\varepsilon_{\text{state}}^{\text{UB}}$ (min-cut upper bound on untraced channels) and a gray-box-access closer $\delta_{\text{act}}^{\text{LB}}$ (conditional-DPI lower bound via admissible probes). The two certificates are independent axes. ReAct controlled replay witnesses near-zero $\delta_{\text{act}}^{\text{LB}} = 0.0163$ bits (95% CI $[0.0124, 0.0208]$) while logging ablation drives $\varepsilon_{\text{state}}^{\text{UB}}$ from $16{,}464$ to 0 bits. Multi-agent counterfactual report intervention witnesses near-saturation at 0.901 bits (CI $[0.873, 0.928]$). LLaDA denoising occupies the mid-spectrum at 0.110 bits (CI $[0.052, 0.234]$), with decision relevance concentrated at the final binding step. A Lean 4 artifact mechanizes Theorem 1 axiom-free from finite-PMF first principles and reduces Propositions 1–2 to four explicit information-theoretic axioms.

1 Introduction

Consider a ReAct-style agent shipped with an unlogged scratchpad of capacity $16{,}384$ bits per turn. A regression test on output traces shows no behavioral change, yet the same trace under controlled replay with the scratchpad disabled reveals a 0.0163-bit shift in the tool-token probability distribution on planning tasks. The disagreement is not a probing artifact. It is a structural feature of the audit: the recorded trace fails to distinguish two qualitatively different systems, one whose hidden capacity is dormant on the test distribution and one whose hidden capacity is active. Identifying which holds is the audit task.

This failure is not a limitation of a particular probe or estimator. It is an identifiability gap. Given any observable joint distribution over traces and actions, there exist two agent realizations—one with zero residual decision relevance, one with maximal—that produce exactly the same observational data.

Contributions.

1. **Theorem 1—Output-Trace Identifiability Gap.** Behavioral equivalence does not imply audit equivalence: two realizations can saturate the $[0, H(A_t \mid \tilde{T}_t)]$ range of δ_{act} while producing identical observational marginals.
2. **Dual-certificate gap-closers.** Proposition 1 (structural-access closer) and Proposition 2 (gray-box-access closer) close the gap via deployment-topology min-cut and admissible probes under conditional DPI. The two certificates are independent axes.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

3. **Three empirical realizations spanning the gap.** ReAct controlled replay provides a near-zero witness ($\delta_{\text{act}}^{\text{LB}} = 0.0163$ bits); LLaDA denoising occupies the mid-spectrum (0.110 bits); multi-agent counterfactual report intervention approaches the binary-action ceiling (0.901 bits). Each instance indexes $\delta_{\text{act}}^{\text{LB}}$ over a different hidden-state coordinate.
4. **Lean 4 mechanization.** Theorem 1 is mechanized axiom-free; Propositions 1–2 are reduced to four explicit information-theoretic axioms (chain rule, cut-set bound, conditional Markov, conditional DPI), with hidden-assumption status verified.

2 Related Work

Probing, patching, and causal abstraction. Causal abstraction [Geiger et al., 2021], latent-knowledge elicitation [Burns et al., 2022], nullspace projection [Ravfogel et al., 2020], and amnesic probing [Elazar et al., 2021] are natural sources of proxy-style evidence: when they expose a probe variable $Z_t = \varphi(S_t)$ and the required conditional assumptions hold, its MI with action can be interpreted through the proxy certificate. Causal tracing/ROME [Meng et al., 2022], representation engineering [Zou et al., 2023], and activation addition [Turner et al., 2023] provide intervention-style evidence when the patch affects action only through S_t.

Black-box audit and network information theory. Casper et al. [Casper et al., 2024] argue that black-box access is insufficient for auditing internal agent mechanisms. Our work makes this insufficiency quantitative for the audit functional δ_{act}: output-only access cannot distinguish $\delta_{\text{act}} = 0$ from $\delta_{\text{act}} = H(A_t \mid \tilde{T}_t)$. Pearl's partial-identification framework [Pearl, 2009] notes that latent-variable models are generally not identifiable from observational data alone; we instantiate this principle for the audit functional δ_{act} and pair it with explicit gap-closers. Property testing [Goldreich et al., 1998] and black-box safety auditing [Su et al., 2024] study what can be inferred from outputs alone. Output-only access can support behavioral tests, but it does not by itself give a lower bound on δ_{act}; $\varepsilon_{\text{state}}^{\text{UB}}$ remains computable from topology when structural access is available. The static certificate proof applies a cut-set upper bound [El Gamal and Kim, 2011] to the time-unrolled DAG (full derivation in Appendix A).

Diffusion language-model agents. LLaDA instantiates large-scale masked diffusion language modeling with bidirectional denoising and instruction-following ability [Nie et al., 2025]. Recent agent work further studies diffusion LMs in multi-step decision and tool-use workflows, including matched comparisons to autoregressive agents [Zhen et al., 2026]. These systems make intermediate denoising latents a natural non-ReAct hidden channel for the dynamic certificate.

3 Setup and Realization Formalism

Standard information-theoretic notation follows Cover and Thomas [2006]. All logarithms are base 2; entropy and mutual information are reported in bits.

At step t: *full trace* T_t (all intermediate activations), *recorded trace* $\tilde{T}_t \subseteq T_t$ (auditor-visible), *operative state* S_t (internal computation), *action* A_t (next token), *unrecorded sources* U_t (exogenous inputs not in $\tilde{T}_t$).

Definition 1 (Agent Realization). *An* agent realization $\mathcal{R} = (P_S, f_T, f_A, \pi_U)$ *consists of an operative-state distribution P_S over S_t, a trace-generation map $f_T : S_t \to \Delta(\tilde{T}_t)$, an action policy $f_A : (S_t, \tilde{T}_t) \to \Delta(A_t)$, and a source policy π_U governing U_t. The realization induces an observational marginal $P_{\mathcal{R}}(\tilde{T}_t, A_t)$ on the support of a fixed deployment prompt distribution P_{prompt}.*

Definition 2 (Behavioral Equivalence). *Two realizations $\mathcal{R}_0, \mathcal{R}_1$ are* behaviorally equivalent*, written $\mathcal{R}_0 \sim_B \mathcal{R}_1$, iff they produce identical joint distributions over observable variables on the deployment prompt distribution:*

$$P_{\mathcal{R}_0}(\tilde{T}_t, A_t) = P_{\mathcal{R}_1}(\tilde{T}_t, A_t) \quad \forall (\tilde{T}_t, A_t) \in \operatorname{supp}(P_{prompt}).$$

Behavioral equivalence concerns the fixed deployment marginal only. Paraphrase variation and red-teaming are generalization probes that explore the policy under distribution shift; they are not audits on the deployment marginal and do not by themselves refute behavioral equivalence.

Symbol	Definition
S_t	Operative state (internal computation at step t)
T_t	Full trace (all intermediate activations)
$\tilde{T}_t$	Recorded trace (auditor-visible subset of T_t)
A_t	Action (next observable token or tool call)
U_t	Unrecorded sources (exogenous inputs not in $\tilde{T}_t$)
$\varepsilon^{\text{UB}}_{\text{state}}$	Static certificate: upper bound on $H(S_t \mid \tilde{T}_t)$
$\delta^{\text{LB}}_{\text{act}}$	Dynamic certificate: lower bound on $I(S_t; A_t \mid \tilde{T}_t)$

Table 1: Notation table.

3.1 Two Core Audit Quantities

Definition 3 (Core Audit Quantities)**.** *The two quantities targeted by the dual certificate are:*

- ***Structural unrecoverability:*** *$\varepsilon_{state} := H(S_t \mid \tilde{T}_t)$, the residual entropy of the operative state given the visible trace.*
- ***Residual decision relevance:*** *$\delta_{act} := I(S_t; A_t \mid \tilde{T}_t)$, the mutual information between operative state and next action given the visible trace.*

The dual-certificate framework in §5 targets upper and lower bounds on these quantities respectively.

4 Output-Trace Identifiability Gap

Theorem 1 (Output-Trace Identifiability Gap)**.** *Let S, T, A be finite. For any finite PMF Q on $T \times A$ with $\sum_a Q(t,a) > 0$ for all $t \in \operatorname{supp}(Q_T)$ and $|S| \geq |A|$, there exist two finite joint PMFs P_0, P_1 on $S \times T \times A$ such that*

(i) P_0 and P_1 project to the same observational marginal Q: $\sum_s P_0(s,t,a) = \sum_s P_1(s,t,a) = Q(t,a)$;

(ii) under the audit functional $\delta_{act}(P) := I_P(S; A \mid T)$, we have $\delta_{act}(P_0) = 0$ and $\delta_{act}(P_1) = H_Q(A \mid T)$.

Hence no functional of the observational marginal Q alone can distinguish a realization with $\delta_{act} = 0$ from one with maximal δ_{act}.

Construction. Define two PMFs on $S \times T \times A$:

$$P_0(s,t,a) = \mathbf{1}[s = s_0] \cdot Q(t,a) \quad \text{(Dirac state: } S \text{ constant, no residual decision relevance)},$$
$$P_1(s,t,a) = \mathbf{1}[s = \phi(a)] \cdot Q(t,a) \quad \text{(state copies action via injection } \phi : A \hookrightarrow S\text{)}.$$

For (i): $\sum_s P_0(s,t,a) = Q(t,a)$ and $\sum_s P_1(s,t,a) = Q(t,a)$ by construction—both project to the same Q. For (ii): In P_0, $S = s_0$ deterministically, so $I_{P_0}(S; A \mid T) = 0$. In P_1, $S = \phi(A)$, so A is determined by S given any T, giving $I_{P_1}(S; A \mid T) = H_Q(A \mid T)$. The cardinality condition $|S| \geq |A|$ guarantees the injection ϕ exists. □

Hidden-assumption verification. Appendix D gives Lean 4 mechanization details.

5 Gap-Closing Certificates

By the data processing inequality, $\delta_{\text{act}} \leq \varepsilon_{\text{state}}$. The two certificates inherit this ordering: if $\varepsilon^{\text{UB}}_{\text{state}} = 0$ then $\delta^{\text{LB}}_{\text{act}} = 0$; the converse does not hold—$\delta^{\text{LB}}_{\text{act}} = 0$ is consistent with large $\varepsilon^{\text{UB}}_{\text{state}}$ (dormant unrecoverable state). Each certificate requires a specific access mode: structural (deployment topology, sufficient for $\varepsilon^{\text{UB}}_{\text{state}}$ via §5.1), controlled replay (re-execution under wild/replay conditions), or proxy/intervention (read $Z_t = f(S_t)$ or perturb a hidden module, supporting conditional-DPI lower bounds via §5.2). The empirical core uses these access modes in a controlled ReAct agent.

5.1 Structural-Access Closer: Static Upper Bound via Untraced-Channel Capacity

The structural-access closer upper-bounds $\varepsilon_{\text{state}} = H(S_t \mid \tilde{T}_t)$ from deployment topology. Its value depends on *which components the auditor trusts*.

Macro-Graph and Trust Partition. The deployment is represented at two levels. At the coarse level, a *macro-graph* partitions components into Trusted ($\mathcal{T}$, protocol-budgeted via interface specs), Untrusted ($\mathcal{U}$, adversarial-budgeted), and Suspect ($\mathcal{S}$, the audit target itself). Each macro-edge expands into a time-unrolled micro-DAG $G_t = (V_t, E_t)$ as below. The trust classification of the source component determines which budget regime applies to every micro-edge in the expansion.

Time-Unrolled Micro-Graph and Dual Budgets. Let $G_t = (V_t, E_t)$ be the time-unrolled DAG of the deployment up to step t. As before, U_t are unrecorded sources, $\tilde{T}_t$ the visible trace, S_t the operative state, and $\mathcal{C}_{\text{unlogged}}$ all reachable cuts separating U_t from S_t.

Each edge $e \in E_t$ carries a per-edge budget that depends on the trust status of its source component:

- **Protocol budget** c_e^{proto}: the maximum information flow under correct protocol execution. Read from interface specifications (e.g., $K \log |\mathcal{V}|$ for a K-token text channel; Appendix A gives the full taxonomy).
- **Adversarial budget** $c_e^{\text{adv}} := \sup_{\mathcal{A} \in \mathfrak{A}(e)} I(X_e^{\mathcal{A}} \to Y_e \mid \tilde{T}_t)$, where $\mathfrak{A}(e)$ is the set of all strategies compatible with the channel's deployment format. By definition $c_e^{\text{adv}} \geq c_e^{\text{proto}}$, with equality only when the channel format constrains the adversary to protocol behavior.

Edges originating in $\mathcal{T}$ use c_e^{proto}; edges crossing into or originating in $\mathcal{U}$ use c_e^{adv}. Appendix F unpacks the continuous spectrum of adversarial budgets for practitioners.

A set of edges is *software-orthogonal* if, conditioned on $\tilde{T}_t$, the channel outputs factor: $p(\{y_e\}_{e \in E'} \mid \{x_e\}, \tilde{T}_t) = \prod_{e \in E'} p(y_e \mid x_e, \tilde{T}_t)$.

Lemma 1 (Additive decomposition)**.** *Under software orthogonality, the induced cut capacity decomposes additively: $C_{cut}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e^{budget(e)}$, where budget$(e)$ = proto for Trusted edges and adv for Untrusted edges.*

When software orthogonality fails, superadditivity of non-orthogonal Untrusted channels is possible (Appendix A discusses remedies).

Proposition 1 (Structural-Access Closer: Static Certificate via Cut-Set Bound)**.** *For any cut Ω separating U_t from S_t with induced capacity $C_{cut}(\Omega) = \sup_{p(X_\Omega \mid \tilde{T}_t)} I(X_\Omega \to Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c})$,*

$$H(S_t \mid \tilde{T}_t) \;\leq\; \underbrace{H(S_t \mid T_t)}_{\varepsilon_{nominal}} \;+\; \min_{\Omega} C_{cut}(\Omega).$$

Setting $\varepsilon_{state}^{UB} := \varepsilon_{nominal} + \min_\Omega C_{cut}(\Omega)$ gives $\varepsilon_{state} \leq \varepsilon_{state}^{UB}$.

Here X_Ω denotes the input signals entering edges crossing the cut, Y_{Ω^c} the corresponding output signals, and X_{Ω^c} the boundary state on the sink side (formal definitions in Appendix A).

Corollary 1 (Additive form)**.** *Under software orthogonality, $\varepsilon_{state}^{UB} = \varepsilon_{nominal} + \min_{C \in \mathcal{C}_{unlogged}} \sum_{e \in C} c_e$, a discrete min-cut on the unlogged subgraph.*

Proof sketch: Chain-rule gives $H(S_t \mid \tilde{T}_t) = H(S_t \mid T_t) + I(S_t; M_t \mid \tilde{T}_t)$ where $M_t = T_t \setminus \tilde{T}_t$. Every path from U_t to S_t crosses a cut Ω, so M_t is d-separated from S_t by the cross-cut signals; DPI yields $I(S_t; M_t \mid \tilde{T}_t) \leq C_{\text{cut}}(\Omega)$. Minimizing over cuts and substituting Lemma 1 completes the proof (full derivation in Appendix A).

Corollary 2 (Autoregressive zero-cut)**.** *If the system is a strict autoregressive core and $\tilde{T}_t$ includes the full context window, then $\mathcal{C}_{unlogged} = \varnothing$, every cut has zero induced capacity, and $\varepsilon_{state}^{UB} = 0$.*

5.2 Gray-Box-Access Closer: Decision Relevance via Conditional DPI

The gray-box-access closer targets $\delta_{\text{act}} := I(S_t; A_t \mid \tilde{T}_t)$. Since S_t is unobservable, we specify admissible probes whose measured variables satisfy $X_t \to S_t \to A_t$ given $\tilde{T}_t$. Conditional DPI supplies the correctness argument: $I(X_t; A_t \mid \tilde{T}_t) \leq \delta_{\text{act}}$.

Proposition 2 (Gray-Box-Access Closer: Probe Certificates from Conditional DPI)**.** *For any admissible probe variable X_t with $X_t \to S_t \to A_t$ given $\tilde{T}_t$, the probe measurement $I(X_t; A_t \mid \tilde{T}_t)$ is a valid lower-bound certificate for δ_{act}.*

Admissible probe: Replay. Let $R_t \in \{\text{wild}, \text{replay}\}$ indicate whether a missing state fragment is reconstructed under the same visible trace $\tilde{T}_t$ (controlled re-execution, not paraphrase variation). Then $I(R_t; A_t \mid \tilde{T}_t) = \text{JS}(P_{\text{wild}}, P_{\text{replay}} \mid \tilde{T}_t) \leq \delta_{\text{act}}$. Discrete action spaces admit direct empirical JS estimation over either realized actions or model-reported action probabilities; the experiments below state which is used.

Admissible probe: Intervention. Let ξ_{hidden} be an exogenous perturbation to a suspected unlogged module (cache, memory buffer, scratchpad). If $\xi_{\text{hidden}} \to S_t \to A_t$ given $\tilde{T}_t$, then $I(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t) \leq \delta_{\text{act}}$. Discrete action spaces admit direct conditional action-distribution shift estimates, such as JS divergence; for continuous action spaces, InfoNCE [van den Oord et al., 2018] or MINE [Belghazi et al., 2018] can be used. Requires gray-box perturbation access; the most direct causal probe of the three.

Admissible probe: Proxy. Let $Z_t = f(S_t)$ be a readable coarsening of the operative state. Since $Z_t \leftarrow S_t \to A_t$ given $\tilde{T}_t$, conditional DPI yields $I(Z_t; A_t \mid \tilde{T}_t) \leq \delta_{\text{act}}$, estimated via the cross-entropy gap of predictors with and without Z_t.

Probe aggregation. A single certificate class can miss some causal paths. Define $\delta_{\text{act}}^{\text{LB}} := \max\{I(R_t; A_t \mid \tilde{T}_t), I(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t), I(Z_t; A_t \mid \tilde{T}_t)\}$. By monotonicity of the maximum, the aggregate remains a valid lower bound on δ_{act}.

Indexed activation profiles. The scalar certificate is the maximum over an indexed family of admissible probes. Keeping the index gives an *activation profile*

$$\Delta(j) := I(X_t^{(j)}; A_t \mid \tilde{T}_t),$$

where j may denote a scratchpad field, a denoising step, an activation layer, or a communication edge. The static certificate identifies where residual capacity can reside; the activation profile identifies where that capacity is behaviorally used. The experiments below keep this algebra fixed while changing the index: module in ReAct, time in the diffusion LM, and communication edge in the multi-agent setting.

6 Empirical Instances Across the Gap

Theorem 1 predicts a spectrum: two behaviorally equivalent realizations can occupy the endpoints 0 and $H(A_t \mid \tilde{T}_t)$ while producing identical observational marginals. We instantiate the framework on three deployments traversing the δ-spectrum: ReAct under controlled replay (low δ, dormant capacity), LLaDA denoising (mid δ, late-binding capacity), and multi-agent private-report intervention (high δ, near-saturated capacity). The static min-cut (§6.1) provides a shared topology coordinate; §§6.2–6.4 traverse the spectrum. Read-only proxy and synthetic ground-truth calibrations are reported in Appendices C and E. All pipelines ran on a single M4 Max workstation.

6.1 Static Certificate: Min-Cut and Logging Ablation

We extract the time-unrolled DAG from Qwen2.5-7B-Instruct ReAct execution traces on tool-selection tasks (Fig. 1, left). Per-edge budgets c_e are assigned from interface specifications (§5.1); the min-cut on unlogged edges gives $\varepsilon_{\text{state}}^{\text{UB}}$. Varying logging levels recomputes the min-cut; Table 2 reports the stepwise reduction from 16,464 to 0 bits. The dominant bottleneck is the scratchpad-read path (16,384 bits).

Logging level	$\varepsilon_{\text{state}}^{\text{UB}}$ (bits)	Bottleneck
ReAct, output only	16,464	scratchpad_read (16384) + router_logits (80)
ReAct, +log router	16,384	scratchpad_read (16384)
ReAct, +log scratchpad	0	none
ReAct, full instrumentation	0	none
LLaDA-8B-Instruct (K=128)	32,768	denoising interface
Multi-agent, all logged	0	none
Multi-agent, peer report unlogged	32,768	peer_report $\to S_t$

Table 2: Static certificate geometry. Adding ReAct logs monotonically removes residual cut capacity; the same min-cut analysis localizes capacity in a scratchpad module, denoising interface, or peer-report edge. The LLaDA bound is independent of K because repeated reuse of the same bounded latent interface does not increase the one-step information bottleneck.

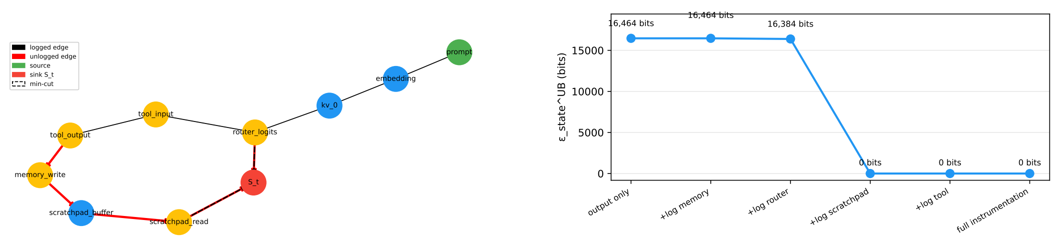

Figure 1: Left: Time-unrolled DAG extracted from Qwen2.5-7B-Instruct ReAct execution (logical edges omitted for clarity). Red edges are unlogged. Dashed edges form the min-cut. Right: Logging ablation curve. Adding instrumentation at the bottleneck drops $\varepsilon_{\text{state}}^{\text{UB}}$ stepwise to zero.

6.2 Dynamic Causal Certificates: Intervention and Controlled Replay

This is the cleanest test of the gap: $\varepsilon_{\text{state}}^{\text{UB}}$ unchanged, $\delta_{\text{act}}^{\text{LB}}$ varies with task—confirming the two certificates are independent axes. We separate the two axes with a dormant/active task split under one topology. The unlogged scratchpad is present in both cases: irrelevant for calculator tasks, necessary for planning tasks. Intervention perturbs the scratchpad (masking or replacement); controlled replay removes it entirely while holding $\tilde{T}_t$ fixed. Table 3 reports JS divergence; for the headline replay contrast, the argmax tool counts are unchanged, so the evidence is a policy-probability shift rather than a realized-tool flip.

The hidden channel is structurally present in both task splits (contributing 16,384 bits to $\varepsilon_{\text{state}}^{\text{UB}}$) but becomes action-relevant only on planning tasks. Appendix C analyses why the read-only proxy variant underperforms; the gap is Pearl's Causal Hierarchy distinction between Level 1 (association) and Level 2 (intervention).

6.3 Mid-Spectrum: LLaDA Denoising Trajectory

LLaDA refines the picture: same dual-certificate framework, but indexed over time rather than module—the activation profile localizes binding to step 10. A diffusion LM provides a temporal mid-spectrum instance. The hidden denoising channel exists throughout the K-step trajectory, but decision relevance concentrates at the final binding step. We use LLaDA-8B-Instruct with K=10 denoising steps on tool-selection prompts. A Gaussian perturbation (σ=5.0) is applied to layer-1 activations at steps $\{2, 4, 6, 8, 10\}$; a late-layer control perturbation tracks broad sensitivity. Each point uses n=20 trajectories and a trajectory-block bootstrap with 1,000 resamples.

The pattern is late-binding: steps 2–8 remain near the floor (0.001–0.006 bits), while step 10 rises to $\delta_{\text{act}}^{\text{LB}} = 0.110$ bits (95% CI $[0.052, 0.234]$). The control layer also rises at step 10 (0.092 bits, CI $[0.043, 0.186]$), so the sensitivity is global. LLaDA carries hidden capacity throughout denoising; it becomes action-relevant only at the discrete tool-choice binding step, placing it mid-spectrum between the ReAct near-zero and multi-agent near-saturation witnesses.

Task	Condition	JS (bits; nats)	95% CI (bits)
	Intervention		
Calculator (dormant)	mask 0.7	0.0000; 0.0000	$[0, 0]$
Planning (active)	mask 0.7	0.0215; 0.0149	$[0.0094, 0.0410]$
Calculator (dormant)	replace 1.0	0.0000; 0.0000	$[0, 0]$
Planning (active)	replace 1.0	0.0055; 0.0038	$[0.0016, 0.0131]$
	Controlled replay		
Calculator (dormant)	scratchpad removed	0.0000; 0.0000	$[0, 0]$
Planning (active)	scratchpad removed	0.0163; 0.0113	$[0.0124, 0.0208]$

Table 3: Dynamic certificates under the same unlogged-scratchpad topology. Dormant calculator tasks show no detectable scratchpad influence; active planning tasks show positive action-distribution shift. The headline replay contrast does not change argmax tool counts.

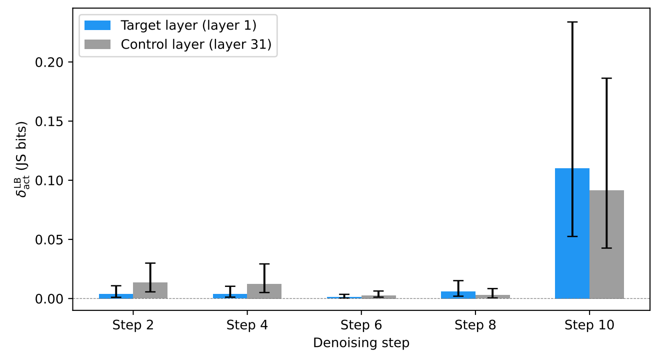

Figure 2: Temporal activation profile for LLaDA-8B-Instruct. Early denoising steps have low dynamic certificates; the final action-forming step has a much larger certificate for both target and control layers, indicating late-binding decision relevance.

6.4 Near-Saturation: Multi-Agent Private-Report Intervention

The multi-agent peer-report channel (Table 2) provides a near-saturation witness: the hidden communication edge carries nearly the full binary-action ceiling. The controller delegates evidence gathering to a worker, so the worker's private report is an epistemic edge in the decision graph. The worker sees a benchmark item and writes a short report; the controller sees only that report and chooses one API from {`calculator`, `search`}. The task pool combines 200 GSM8K arithmetic items [Cobbe et al., 2021] and 200 HotpotQA multi-hop fact questions [Yang et al., 2018] with seed 42.

The main intervention replaces the worker report with a report from the opposite task class while holding the controller protocol fixed. A neutral replay removes the report content, and a same-class shuffle replaces it with another report from the same task class. An oracle-tag condition directly names and flips the API label to measure the binary ceiling. For each task and condition, the controller is sampled $k{=}8$ times; labels are parsed as {`calculator`, `search`, `invalid`}; and $\mathrm{JS}(P_{\text{base}}, P_{\text{condition}})$ is averaged with a task-block bootstrap.

Reclassifying the worker from Trusted to Untrusted shifts the audit strategy from detection to mandatory topology redesign, even though the topology and dynamic certificate ($\delta_{\text{act}}^{\text{LB}} = 0.901$ bits) are identical: under a Trusted protocol budget the worker report contributes 32,768 bits to $\varepsilon_{\text{state}}^{\text{UB}}$ and monitoring suffices, but under an adversarial budget the channel requires logging, capacity restriction, or cryptographic commitment. Trust misclassification is unidirectional—mistaking Untrusted for Trusted underestimates $\varepsilon_{\text{state}}^{\text{UB}}$, while the reverse is conservative.

Condition	$\delta_{\text{act}}^{\text{LB}}$ (JS bits)	95% CI (bits)
Neutral replay	0.358	$[0.324,\ 0.392]$
Counterfactual evidence	0.901	$[0.873,\ 0.928]$
Same-class shuffle	0.084	$[0.060,\ 0.109]$
Oracle tag upper-bound	1.000	$[1.000,\ 1.000]$

Table 4: Dynamic certificates for the multi-agent deployment. Counterfactual evidence nearly saturates the binary action ceiling, while same-class shuffling remains much smaller.

7 Discussion

The two-evidence-types frame. Existing audit frameworks collapse hiddenness into a scalar; our results require two structurally independent kinds of evidence (topology, behavior) and a third axis (trust) for action. This decomposition is what makes the gap closable rather than merely diagnosable.

Limitations. Behavioral equivalence (Definition 2) is defined with respect to a fixed deployment prompt distribution P_{prompt}; distribution shift under red-teaming or deployment drift is not covered. The structural-access closer (Proposition 1) requires software orthogonality for tightness of the per-edge additive decomposition; shared hidden state across unlogged channels weakens the min-cut interpretation without invalidating the upper bound (Appendix A). The gray-box-access closer (Proposition 2) requires the probe to satisfy the conditional Markov chain $X_t \to S_t \to A_t$ given $\tilde{T}_t$; probe misspecification can underestimate δ_{act}.

Trust boundaries in practice. The trust partition $(\mathcal{T}, \mathcal{U}, \mathcal{S})$ is an auditor-side modeling choice, not a property of the deployment. The multi-agent experiment shows that varying only the trust assumption—without changing topology or dynamic data—can shift the audit strategy from detection to mandatory redesign. This has direct implications for AI supply-chain auditing: when a deployment integrates a third-party model, API, or retrieval module, the auditor must decide whether to treat that component as Trusted (protocol-budgeted, audited via $\delta_{\text{act}}^{\text{LB}}$) or Untrusted (adversarial-budgeted, requiring topology-level mitigation). The default-deny principle applies: absent cryptographic or organizational trust anchors, components should be classified as Untrusted to avoid systematic underestimation of $\varepsilon_{\text{state}}^{\text{UB}}$.

Future Directions. Future work should lift the finite-support assumption to continuous state spaces, automate dynamic-certificate probe synthesis via active probe selection rather than hand design, and connect the Lean artifact to a verified audit toolchain that emits machine-checked certificates per deployment.

Acknowledgments

We thank the developers of Qwen2.5 and LLaDA for the open-weight models used in the ReAct and diffusion-LM experiments; the creators and maintainers of GSM8K and HotpotQA for benchmark data used in the multi-agent setting; and the Hugging Face Transformers, PyTorch, NumPy, scikit-learn, NetworkX, Matplotlib, Lean 4, Lake, and Mathlib communities for the software infrastructure supporting the experiments, figures, and formalization. Computations ran on a single Apple M4 Max workstation.

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NeurIPS Paper Checklist

1. Claims Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: Yes. The abstract and introduction state the identifiability gap (Theorem 1), the dual-certificate gap-closers (Propositions 1–2), and the three empirical realizations spanning the gap. All claims are matched by theoretical derivations (§4–5) and empirical evidence (§6).

2. Limitations Does the paper discuss limitations of the work?

Answer: Yes. Limitations are discussed in several places: the observational bottleneck and proxy-estimator bias/variance (Appendix C), the scope of claims (§7), the conservative nature of the static bound, and the dependence on software-orthogonality assumptions for tightness (§5.1, Appendix A).

3. Theory Assumptions and Proofs Did you state the full set of assumptions of all theoretical results, and did you include complete proofs of all theoretical results?

Answer: Yes. All assumptions are stated with each result: the cardinality condition $|S| \geq |A|$ for Theorem 1, software orthogonality for Lemma 1, and conditional Markov/DPI for Proposition 2. Complete proofs are given in the main text (Theorem 1 construction proof, proof sketches for Propositions 1–2) and in full detail in Appendix A. A Lean 4 artifact mechanizes Theorem 1 axiom-free and reduces Propositions 1–2 to four explicit axioms (Appendix D).

4. Experimental Result Reproducibility Did you provide a way to reproduce the experimental results?

Answer: Yes. Detailed experimental protocols are described in Appendix B, including the static-certificate computation procedure and the dynamic-certificate estimation steps (proxy, replay, intervention, and private-report intervention). The paper references implementation README files. Code, data, and the Lean artifact are included in the supplementary material.

5. Open access to data and code Did you include the code, data, and instructions needed to reproduce the main experimental results?

Answer: Yes. Code, precomputed results, and reproduction instructions are provided in the supplementary material. The experiments use the open-weight Qwen2.5-7B-Instruct model (Apache 2.0).

6. Experimental Setting/Details Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)?

Answer: Yes. Experimental details are reported in §6 and Appendix B: e.g., 5-fold cross-fitted logistic regression, trajectory-block bootstrap with 95% CIs, PCA dimensions, audit discretization $Q = 256$, perturbation levels, controller sample count k, and task-pool sampling seed. Hyperparameter choices are motivated and described.

7. Experiment Statistical Significance Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: Yes. All experimental tables report 95% confidence intervals obtained via trajectory- or task-block bootstrap (§6.2–6.4, Appendix C). Error bars are shown in the appendix proxy figure (Figure 3). The method for calculating CIs is explained in the text and Appendix B.

8. Experiments Compute Resource For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: Yes. The paper states that all pipelines ran on a single Apple M4 Max workstation (§6). This indicates the hardware class and scale; no GPU-cluster requirements are needed.

9. Code Of Ethics Have you read the NeurIPS Code of Ethics and ensured that your research conforms to it?

Answer: Yes. The authors have read and adhere to the NeurIPS Code of Ethics. The work focuses on safety-oriented auditing of AI agents and does not involve human subjects, privacy-sensitive data, or dual-use risks beyond those discussed in the paper.

10. Broader Impacts If appropriate for the scope and focus of your paper, did you discuss potential negative societal impacts of your work?

Answer: Yes. The paper develops a framework for auditing AI agents, aimed at improving transparency and safety. Potential dual-use concerns (e.g., adversaries studying the certificates to design harder-to-audit systems) are acknowledged in the limitations and the conservative nature of the bounds (§7). The framework is inherently defensive in scope.

11. Safeguards Do you have safeguards in place for responsible release of models with a high risk for misuse (e.g., pretrained language models)?

Answer: N/A. The paper does not release any new pretrained models. It uses existing open-weight models (Qwen2.5) as-is for auditing experiments without modifying or redistributing them.

12. Licenses Did you cite the creators and respect the license and terms of use of existing assets?

Answer: Yes. The paper cites the model and dataset assets where applicable, acknowledges the supporting software communities (Qwen2.5, LLaDA, Hugging Face Transformers, PyTorch, NumPy, scikit-learn, NetworkX, Matplotlib, Lean 4, Lake, and Mathlib; §*Acknowledgments), and uses publicly available assets under their respective licenses.

13. New Assets Are you releasing new assets (datasets, code, models), and did you document them properly?

Answer: No. The experimental code, data, and Lean 4 formalization are provided in the supplementary material for review and will be publicly released and documented upon acceptance.

14. Crowdsourcing and Research with Human Subjects Did you use crowdsourcing or conduct research with human subjects?

Answer: N/A. The research does not involve crowdsourcing or human subjects.

15. IRB Approvals Did you describe potential participant risks and obtain Institutional Review Board (IRB) approvals (or an equivalent approval/review)?

Answer: N/A. No human subjects research was conducted; therefore, IRB approval is not applicable.

16. Declaration of LLM usage Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: N/A. The LLM (Qwen2.5) is only the audit target—not a component of the auditing method. The framework (min-cut computation, mutual-information lower bounds, dual-certificate logic) is independent of any particular language model.

A Network Information Theory Background for Proposition 1

Conditional channel capacity. For each unlogged edge e in the time-unrolled DAG G_t, the conditional capacity c_e is the maximum information that can flow through e under the worst-case distribution consistent with the system protocol. It is not an empirical average mutual information. It is a structural upper bound read from interface specifications: token window length, hidden-dimension size, quantization level, and discrete-state cardinality. A text channel with budget K tokens over vocabulary $\mathcal{V}$ has $c_e \leq K \log |\mathcal{V}|$; a vector state of dimension d at quantization Q has $c_e \leq d \log Q$; a discrete scheduler with $|\Omega|$ states has $c_e \leq \log |\Omega|$. The $K \log |\mathcal{V}|$ form is the maximum-entropy bound (uniform over K-token sequences); any other distribution gives strictly lower entropy. Auditors who can measure the empirical input distribution to a particular channel may substitute a tighter bound, which only improves the certificate.

What tightening means. The static certificate remains valid under any replacement $c_e^\star \geq I(X_e; Y_e \mid \tilde{T}_t)$, so tightening is a matter of choosing smaller but still defensible per-edge budgets. In practice, this means replacing the worst-case $K \log |\mathcal{V}|$ or $d \log Q$ budget with an empirical or conditional estimate at the same audit discretization, rather than changing the theorem itself.

Cut-set upper bound and additive decomposition. Let $G_t = (V_t, E_t)$ be the time-unrolled DAG. For a cut $\Omega \subseteq V_t$ separating source nodes from a target node, the *induced cut capacity* $C_{\text{cut}}(\Omega)$ is the supremum, over joint input distributions $p(X_\Omega \mid \tilde{T}_t)$, of the directed conditional mutual information $I(X_\Omega \to Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c})$. The network-information-theoretic cut-set upper bound [El Gamal and Kim, 2011] states that any information flowing from sources in Ω to a target in Ω^c is bounded above by $C_{\text{cut}}(\Omega)$, and minimizing over cuts gives the tightest such bound. Proposition 1 applies this bound to the unrecorded trace fragment M_t. We use the cut-set *upper bound* direction only; the network-coding achievability results [Ahlswede et al., 2000], which concern lower bounds on capacity under cooperative coding across the cut, are not invoked. The induced capacity $C_{\text{cut}}(\Omega)$ is in general a supremum over joint input distributions and cannot be read directly from interface specifications. Lemma 1 shows that under software orthogonality, when the channel laws across the cut factor into independent single-edge channels conditional on $\tilde{T}_t$, the induced capacity is bounded above by the sum of per-edge capacities:

$$C_{\text{cut}}(\Omega) \;\leq \sum_{e \in E(\Omega, \Omega^c)} c_e.$$

Corollary 1 combines this with Proposition 1 to reduce the certificate computation to a discrete minimum-cut problem on G_t with edge capacities c_e. This is the form used in the static certificate experiment and in the protocol in §B.

Formal derivation of Proposition 1. The proof in §5.1 has two steps: a trace-gap identity and a cut-set bound via d-separation. We restate the argument here with the d-separation step explicit.

Step 1: Trace-gap identity. Let $M_t := T_t \setminus \tilde{T}_t$. The chain rule of conditional entropy gives

$$H(S_t \mid \tilde{T}_t) = H(S_t \mid T_t) + I(S_t; M_t \mid \tilde{T}_t).$$

The first term is $\varepsilon_{\text{nominal}} = H(S_t \mid T_t)$.

Step 2: Cut-set bound via d-separation. Fix a cut $\Omega \in \mathrm{Cuts}(U_t \to S_t)$. By the definition of a cut in the time-unrolled DAG G_t, every directed path from a node in Ω to $S_t \in \Omega^c$ must cross at least one edge in $E(\Omega, \Omega^c)$. The causal influence of M_t on S_t must be mediated by the cross-cut message stream $(X_\Omega \to Y_{\Omega^c})$ together with the boundary state X_{Ω^c}. Formally, M_t is d-separated from S_t by $(X_\Omega \to Y_{\Omega^c},\, X_{\Omega^c})$ conditional on $\tilde{T}_t$, which yields

$$I(S_t; M_t \mid \tilde{T}_t) \;\leq\; I\Big(X_\Omega \to Y_{\Omega^c} \,\Big|\, \tilde{T}_t,\, X_{\Omega^c}\Big).$$

By the definition of $C_{\text{cut}}(\Omega)$ as the supremum of the right-hand side over admissible joint input distributions on X_Ω,

$$I(S_t; M_t \mid \tilde{T}_t) \;\leq\; C_{\text{cut}}(\Omega).$$

Since this holds for every cut, it holds for the minimum,

$$I(S_t; M_t \mid \tilde{T}_t) \le \min_{\Omega \in \mathrm{Cuts}(U_t \to S_t)} C_{\text{cut}}(\Omega).$$

Combining with Step 1 completes the proof of Proposition 1. Corollary 1 follows immediately by substituting Lemma 1 into each software-orthogonal cut: under orthogonality, $C_{\text{cut}}(\Omega) \le \sum_{e \in E(\Omega, \Omega^c)} c_e$, and the minimum over cuts becomes a discrete min-cut problem on the unlogged subgraph with edge capacities c_e.

B Experimental Protocols

B.1 Static Certificate Computation Protocol

Inputs: deployment architecture specification, logging manifest, and protocol budgets (token window sizes, hidden dimensions, quantization levels). Procedure: (1) construct the time-unrolled DAG G_t; (2) partition edges into logged and unlogged; (3) verify software orthogonality (Lemma 1) or add shared-state edges explicitly to G_t until orthogonality holds; (4) assign capacity c_e per unlogged edge using the forms in §5.1; (5) enumerate reachable cut sets $\mathcal{C}_{\text{unlogged}}$ with a min-cut algorithm after logged edges are collapsed to zero residual capacity; (6) compute $\varepsilon_{\text{state}}^{\text{UB}} = \varepsilon_{\text{nominal}} + \min_C \sum_{e \in C} c_e$ via Corollary 1.

For the logging ablation, repeat steps (2)–(6) for the ordered manifests used in §6.1: output-only trace, tool-call trace, router-logit trace, summary-buffer trace, and full scratchpad trace. Report the upper-bound value and active minimum-cut edges at each level. See `experiments/7.1_static_certificate/README.md` for implementation.

B.2 Dynamic Certificate Estimation Protocol

Proxy certificate. Inputs: model, task environment, and probe family $Z_t^{(k)} = f_k(S_t)$. Steps: (1) run inference on N trajectories and record $(\tilde{T}_t, Z_t^{(k)}, A_t)$; (2) train cross-fitted predictors $p_\theta(A_t \mid \tilde{T}_t)$ and $p_\phi(A_t \mid \tilde{T}_t, Z_t^{(k)})$; (3) compute the cross-entropy difference $\hat{I}_k = L_{\text{trace}} - L_{\text{trace}+Z^{(k)}}$; (4) estimate trajectory-block bootstrap intervals; (5) repeat for random, label-permuted, 1-dimensional, 3-dimensional, 5-dimensional, and full-slice proxies.

Optional controlled replay certificate. Inputs: model and two controlled execution regimes (wild and replay). Steps: (1) for each $\tilde{T}_t$, run the same system under wild and replayed missing-state conditions; (2) estimate $\hat{\Delta} = \widehat{\mathrm{JS}}(P_{\text{wild}}, P_{\text{replay}} \mid \tilde{T}_t)$; (3) aggregate over trace slices. This is not output-only paraphrase replay; it is valid only under the controlled replay condition in §5.2.

Intervention certificate. Inputs: model, targeted hidden module, perturbation budget, and task split. Steps: (1) inject ξ_{hidden} at levels $\{0, \sigma_1, \sigma_2, \ldots\}$ while holding $\tilde{T}_t$ fixed; (2) record $(\xi_{\text{hidden}}, A_t \mid \tilde{T}_t)$ pairs; (3) estimate $\hat{I}(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t)$ via conditional action-distribution shift for discrete action spaces, or via InfoNCE/MINE for continuous action spaces; (4) repeat on dormant tasks and active tasks under the same topology to test whether $\delta_{\text{act}}^{\text{LB}}$ changes while $\varepsilon_{\text{state}}^{\text{UB}}$ remains fixed.

Multi-agent private-report certificate. Inputs: task pool, worker model, controller model, and a two-class API space. Steps: (1) sample $N/2$ arithmetic tasks and $N/2$ factual-retrieval tasks; (2) have the worker generate a private evidence report under a prompt that forbids direct API names and audit each report for leakage; (3) sample the controller action distribution k times under wild, neutral replay, opposite-class report, same-class shuffled report, and oracle-tag conditions; (4) parse actions into {`calculator`, `search`, `invalid`}; (5) compute per-task JS divergence in bits and task-block bootstrap intervals. The oracle-tag contrast is reported only as an upper bound; the main contrast is the opposite-class worker-report intervention.

See the README files for experiments 7.2, 7.3, and 7.6 for implementation.

C Proxy Estimator: Bias and Variance

The read-only proxy experiment is a calibration of observational access, not a main activation result. On 3,000 tool-selection trajectories (mixed calculator/search/email/calendar/weather), we record $\tilde{T}_t$ (embedding-layer representation), Z_t (layer-24 tool-logit projection), and A_t. The CE-diff $\hat{I} = \mathcal{L}(A_t \mid \tilde{T}_t) - \mathcal{L}(A_t \mid \tilde{T}_t, Z_t)$ is estimated via 5-fold cross-fitted logistic regression with trajectory-block bootstrap. Table 5 reports the resolution ablation: random, 1/3/5-dim PCA, and full 10-dim proxy.

The d=5 PCA proxy recovers 0.0317 bits (1.4% of the $\log_2 5 = 2.32$ bit ceiling) with CI excluding zero on the mixed-task distribution. The magnitude reflects an *observational bottleneck*: when the baseline predictor achieves near-perfect accuracy, the ceiling $H(A_t \mid \tilde{T}_t) \approx 0$ caps any observational proxy. Pearl's Causal Hierarchy [Pearl, 2009] explains the pattern: proxy certificates operate at Level 1 (association), while the intervention and replay certificates in §6.2 operate at Level 2 (intervention), breaking the deterministic data-generating process.

The dormant/active proxy split is an estimator-noise diagnostic. On calculator tasks (dormant scratchpad), $H(A_t \mid \tilde{T}_t) = 0$ empirically, yielding $\delta^{\text{LB}}_{\text{act}} = 0$ trivially. On planning tasks (active scratchpad), the proxy row is positive in point estimate but has a wide CI that straddles zero. We do not use the read-only proxy as the main body evidence for active scratchpad use; the causal intervention and controlled replay in §6.2 supply that evidence.

Proxy	CE-diff (bits; nats)	95% CI (bits)
Resolution ablation (mixed task)		
random	-0.0007; -0.0005	$[-0.0012, -0.0003]$
$d = 1$ PCA	$+0.0065$; $+0.0045$	$[+0.0053, +0.0078]$
$d = 3$ PCA	$+0.0250$; $+0.0173$	$[+0.0225, +0.0273]$
$d = 5$ PCA	$+0.0317$; $+0.0220$	$[+0.0273, +0.0358]$
full (10-dim)	$+0.0183$; $+0.0127$	$[+0.0134, +0.0229]$
Dormant/active split (d=64 PCA, L2-regularized)		
calculator (dormant)	0.0000; 0.0000	$[0, 0]$
planning (active)	$+0.0412$; $+0.0286$	$[-0.0725, +0.1386]$

Table 5: Read-only proxy calibration. Top: resolution ablation on the mixed tool-selection dataset. Bottom: dormant/active split. The active-task row is reported honestly as an estimator-noise diagnostic because its confidence interval straddles zero; causal replay/intervention supplies the body evidence for active scratchpad use. Negative random-control CE-diff is a finite-sample artifact near the noise floor and is clipped to zero when interpreted as a lower-bound certificate.

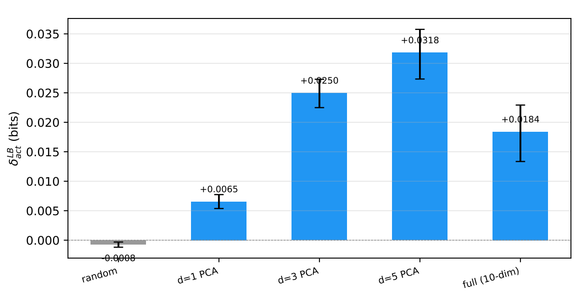

Figure 3: Proxy resolution ablation. $\delta^{\text{LB}}_{\text{act}}$ increases monotonically through d=5 PCA. The random proxy sits near zero (estimator leakage check). Error bars are 95% trajectory-block bootstrap CIs.

The CE-diff estimator $\hat{I} = L(\tilde{T}_t) - L(\tilde{T}_t, Z_t)$ lower-bounds $I(Z_t; A_t \mid \tilde{T}_t)$, but three bias sources and two variance sources matter in practice.

Bias sources. (1) *Proxy coarsening:* $Z_t = f(S_t)$ discards information in S_t not captured by f, so $I(Z_t; A_t \mid \tilde{T}_t) \leq I(S_t; A_t \mid \tilde{T}_t)$. The gap is the coarsening loss. (2) *Restricted predictor class:*

Logistic regression or shallow MLPs may not reach the Bayes-optimal predictor, so the CE-diff underestimates the true conditional MI. (3) *Finite-sample bias:* With finite N, the cross-entropy on held-out folds is an upward-biased estimate of the population CE, and the difference of two such estimates carries residual bias. Cross-fitting mitigates but does not eliminate this.

Variance sources. (1) *Trajectory sampling:* The N observed trajectories are a finite sample from the deployment distribution. Block bootstrap over trajectories estimates the resulting CI. (2) *Proxy dimension:* Higher-dimensional proxies carry more information but increase predictor variance. The PCA ablation in Table 5 shows this trade-off: d=5 outperforms both d=3 and the full 10-dimensional proxy.

Mean squared error. Decomposing MSE $= (\text{bias})^2 +$ variance, the proxy certificate is a conservative lower bound: positive bias (underestimation) is safe for the audit (it never overstates δ_{act}), while variance widens the CI but does not shift the point estimate.

Sampling noise and temperature. Increasing decoding temperature can raise $H(A_t \mid \tilde{T}_t)$, but independent sampler noise is not a certificate amplifier. It changes the audited policy and can reduce $I(Z_t; A_t \mid \tilde{T}_t)$ when the added action variation is unrelated to the hidden state proxy. A temperature sweep is therefore a calibration of a randomized deployment policy, not a substitute for proxy quality or causal access. We do not use temperature injection for the main proxy certificate; the synthetic experiment instead injects hidden-state stochasticity with known ground truth.

D Lean 4 Formalization Boundary

The Lean 4 artifact (`IdentifiabilityGap.lean`) mechanizes Theorem 1 axiom-free from finite-PMF first principles. The existing artifacts (`Screenability.lean`, `InternalImpossibility.lean`) provide machine-verified instances of the autoregressive zero-cut special case of Proposition 1 (Corollary 2). The file `DualCertificate.lean` formalizes the two gap-closing structural reductions corresponding to the static-certificate cut-sum bound and the conditional-DPI dynamic certificate.

The artifact uses an axiomatic interface for measure-theoretic information theory: Theorem 1 is axiom-free; the structural reductions of Propositions 1–2 are machine-checked, while the underlying chain-rule, cut-set, and DPI facts are imported as `axioms` and listed in §D.1.

Main-Text Item	Lean File	Lean Theorem	Status
Thm 1 (Identifiability Gap)	IdentifiabilityGap	`identifiability_gap_extremes`	C
Cor 2 (Autoregressive zero-cut)	Screenability	`no_eis_autoregressive`	C
Prop 1 (Structural-access closer)	DualCertificate	`prop1_static_ub`	C/E
Prop 2 (Gray-box-access closer)	DualCertificate	`prop2_dynamic_lb`	C/E

Table 6: C = machine-checked from Mathlib first principles (Theorem 1 is axiom-free). C/E = structural reduction is machine-checked conditional on the four axioms listed below. Entropy, conditional entropy, and conditional mutual information are defined by finite-discrete formulas using Mathlib finite sums and real logarithms; the four axioms in §D.1 remain external.

D.1 External / Assumed Steps

The following are imported as explicit Lean `axioms` and consumed by downstream machine-checked proofs:

- **For Prop 1:** `chain_rule` (trace-gap chain rule of conditional entropy across the visible/full-trace decomposition); `cut_set_bound` (network-information-theoretic cut-set inequality on the time-unrolled DAG).
- **For Prop 2:** `condMarkov` (conditional Markov property of probe variables); `cond_dpi` (conditional Data Processing Inequality).

The artifact verifies a structural reduction: that the main-text propositions follow from these axioms by elementary algebraic manipulation. It does not derive the axioms themselves from Lean's measure-theoretic foundations.

E Synthetic Ground-Truth Validation

To test whether the certificates recover known hidden influence, we construct a synthetic agent whose hidden-state entropy and decision relevance are controlled by design. A binary hidden variable $H \sim \text{Bern}(0.5)$ injects influence of strength β_h into the action logits; the unlogged-channel capacity $c_H = H(H) = 1$ bit is known. We run both the static min-cut ($\varepsilon^{\text{UB}}_{\text{state}} = 1$ bit) and the proxy CE-diff estimator on $1{,}000$ trajectories at five influence levels.

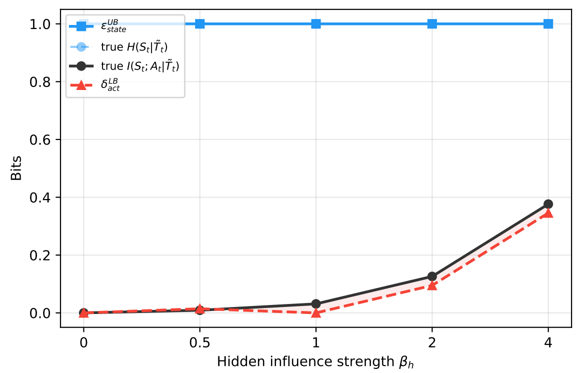

Figure 4: Synthetic ground-truth. $\varepsilon^{\text{UB}}_{\text{state}}$ (blue) equals the true $H(S_t \mid \tilde{T}_t) = 1$ bit. $\delta^{\text{LB}}_{\text{act}}$ (red) tracks the true $I(S_t; A_t \mid \tilde{T}_t)$ from below, with negative values at low β_h reflecting estimator variance near the noise floor.

Fig. 4 shows $\varepsilon^{\text{UB}}_{\text{state}} = 1$ bit matches the true hidden entropy (the bound is tight when there is no redundant unlogged capacity). The dynamic lower bound $\delta^{\text{LB}}_{\text{act}}$ tracks the true conditional MI from below at all tested influence levels ($\beta_h \geq 1$); negative CE-diff values at $\beta_h = 0$ reflect finite-sample variance around a near-zero true MI.

The Qwen deployment has a near-deterministic baseline that suppresses the proxy signal ($H(A_t \mid \tilde{T}_t) \approx 0$). The synthetic environment instead injects controlled stochasticity ($H = 1$ bit of hidden-state entropy). Without that observational bottleneck, the proxy certificate $\delta^{\text{LB}}_{\text{act}}$ tracks the true conditional-MI curve from below in the tested range. The validation shows that the proxy certificate can become tight when structural capacity is behaviorally active *and* the observational ceiling permits detection.

F Adversarial Budget Ladders

The adversarial budget $c^{\text{adv}}_e := \sup_{\mathcal{A} \in \mathfrak{A}(e)} I(X^{\mathcal{A}}_e \to Y_e \mid \tilde{T}_t)$ in §5.1 is defined over the set $\mathfrak{A}(e)$ of all strategies compatible with the channel's deployment format. Restricting $\mathfrak{A}(e)$ yields a ladder of tightening budgets for practical audit:

1. **Protocol-compliant** ($\mathfrak{A}$ restricted to correct protocol execution): $c^{\text{adv}}_e = c^{\text{proto}}_e$. The adversarial budget collapses to the interface specification. Appropriate when the component is trusted by organizational or cryptographic anchor.
2. **Format-compatible** ($\mathfrak{A}$ unrestricted within channel syntax): $c^{\text{adv}}_e = c^{\text{physical}}_e$, the Shannon capacity of the raw channel. This is the default budget when a component is classified as Untrusted without further constraint.
3. **Arbitrary** ($\mathfrak{A}$ unconstrained): $c^{\text{adv}}_e = \infty$. Audit is impossible; the deployment must be redesigned to eliminate or constrain the channel.

In practice, auditors can tighten the certificate by moving down this ladder: imposing format restrictions on a text channel (e.g., structured JSON with a fixed schema) reduces c_e^{physical} below the free-text capacity; requiring cryptographic commitments (e.g., the worker publishes a hash of its report before the controller acts) can drive c_e^{adv} toward zero even for an Untrusted component. Each step corresponds to a concrete engineering intervention, making the adversarial budget ladder a direct bridge between the information-theoretic certificate and deployment-hardening decisions.

For channels shared between multiple Untrusted components, the joint adversarial capacity may exceed the sum of per-edge budgets (superadditivity; §5.1). The remedy is to merge such edges into a single channel family and budget their joint capacity, or to decompose the format into orthogonal sub-channels that provably cannot be coordinated.